Fiscal Year:	FY 2016	**Task Last Updated:**	FY 07/13/2015
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Constrained Vapor Bubble Flight Experiment II – Binary Mixtures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wayner, Peter (Rensselaer Polytechnic Institute)		
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The thermophysical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, related passive engineering systems like heat pipes have not been optimized and it is unclear whether recent advances in the design of these systems based on ground-based research using fluid mixtures can be translated to microgravity systems. Herein, we propose basic experimental and theoretical studies of the nonisothermal constrained vapor bubble (CVB-2) using a mixture of pentane and isohexane under microgravity conditions to help remedy this undesirable situation. This proposed study is multi-faceted: 1) it is a basic scientific study in interfacial phenomena, microgravity fluid physics, and thermodynamics; 2) it is a basic study in thermal and mass transport; and 3) it is a study of a passive heat exchanger. Although the basic scientific Facets (1) and (2) are emphasized, the research naturally leads to the development of a passive heat exchanger optimized for microgravity conditions.
The proposed optical study of a vapor bubble constrained in a transparent glass cell under nonisothermal conditions will

Task Description:	increase the basic understanding of transport systems controlled by interfacial phenomena. The pressure gradient field will be obtained optically whereas the temperature field will be obtained using thermal sensors. Due to the sensitivity of systems of the proposed relatively large size to gravity, to small temperature and pressure gradients, and to changes in fluid composition, these transport systems need to be studied under microgravity conditions. An "axisymmetric" system with a small Bond number is required. Direct extensions to the microgravity environment first investigated in CVB-1 are proposed. The augmented Young-Laplace equation and the Kelvin-Clapeyron model will be used to help understand and quantify how the system operates. The immediate macroscopic objectives of CVB-2 are to determine the stability, the fluid flow characteristics, the average heat transfer coefficient in the evaporator and condenser sections, and the heat conductance of the CVB as a function of the heat input using the new fluid mixture. The immediate microscopic objective is to determine the detailed characteristics of the transport processes in the curved liquid film. The local conditions under which cavitation and instability occur with the formation of a dry region will be determined as a function of heat flux, film thickness, and stress. High fluxes and relatively large systems with small capillary pressures will be emphasized. The data using the pentane/isohexane mixture will be compared with the pure pentane data obtained in CVB-1 and ground-based data on identical fluid mixtures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions regarding phase-change heat and mass transfer operations that are becoming increasingly important in a wide variety of fields ranging from energy conversion, to microelectronics cooling, micro-electromechanical systems (MEMS) devices, self-assembly, and space flight. Of particular concern to NASA are microchannel and heat pipe systems used for thermal management and perhaps separation processes. Such systems are attractive for long-term space flights because they generally require no moving parts and are potentially maintenance free. The ability to control transport processes in an extended meniscus will be of great utility both on Earth and in space, whether we are looking for enhanced heat transfer, coating, or self-assembly applications. Some of our preliminary ground-based experiments concerning the microscopic evaporating meniscus, performed as part of the existing CVB-1 flight experiment program, have indicated that using a fluid mixture, with the minority component being the less volatile, higher surface tension liquid, offers the opportunity to control fluid flow and heat transfer in an evaporating meniscus and thus enhance the performance of heat pipes. The surface concentration gradient will also enhance axial fluid flow in the corners. The extension of the CVB-1 experiment to mixtures allows us to isolate the Marangoni effects from the effects of gravity and begin to understand how to exploit Marangoni flows for improving the performance of heat pipes in space and on Earth.
Task Progress:	During the past year of the research, we have been vigorously working on analyzing the data from the CVB2 and CVB1 flight experiments, writing papers, and presenting the information at as many conferences as we could reasonably attend. The highlights are listed below and the reference details are given in the Bibliography section below. 1. We have spent much of the time this year regressing and analyzing the data from the high heat input experiments run aboard the International Space Station (ISS) in late 2010 using the 30 and 40 mm module experiments on the ISS and developing enhanced thermal models to determine the internal heat transfer rates within the cuvette. This has resulted in one publication in the International Journal of Heat and Mass Transfer and another in Langmuir. 2. We have analyzed the high heat input experiments and have shown that the decrease in performance of the heat pipe is not related to dryout or boiling limitations but instead results from flooding of the hot end. This has been a result never before seen in heat pipes and results from a competition between Marangoni at the hot end and capillary forces at the cold end. This resulted in one publication in Physics Review Letters that received quite a lot of press attention with the article being highlighted in Nature, Science, Scientific American, and Today's Science. 3. We have been compiling and analyzing data from the flight experiments associated with CVB2. We obtained some very interesting results that we thought were completely different from CVB1, but there are some interesting similarities between CVB2 and the 20 mm CVB1 module that we are currently exploring. In addition, it turns out that CVB2, when it was working reliably, performed as expected. Heat transfer coefficients in the evaporator section were nearly double those of CVB1. 4. We presented results at 6 meetings this year. 5. The group received a 2015 ISS R&D Compelling Results award at the Boston Conference in July 2015 for the experimental results associated with the Physical Review Letters paper.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	Kundan A, Plawsky JL, Wayner PC Jr. "Effect of capillary and Marangoni forces on transport phenomena in microgravity." Langmuir. 2015 Apr;31(19):5377-86. http://dx.doi.org/10.1021/acs.langmuir.5b00428 , Apr-2015
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Awards	Kundan A, Wayner PC Jr, Plawsky JL. "2015 ISS R&D Most Compelling Science Results Award, ISS Research and Development Conference in Boston, July 7, 2015." Jul-2015
Significant Media Coverage	Kundan A, Wayner PC Jr, Plawsky JL. "Research Highlight of article 'Thermocapillary phenomena and performance limitations of a wickless heat pipe in microgravity.' Phys Rev Lett. 2015 Apr 10;114(14):146105. http://dx.doi.org/10.1103/PhysRevLett.114.146105 " Infobase Learning online, June 2015., Jun-2015
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Significant Media Coverage	Grocholski B. (Kundan A, Plawsky JL, Wayner PC Jr, et al.) "Uncool Heat Pipes in Microgravity. A synopsis of findings from the CVB flight experiment was reported in the EDITOR'S CHOICE, section of 22 May 2015 issue of the journal Science, p. 877 under the Fluid Dynamics heading." Science. 2015 May 22;348(6237):877. http://dx.doi.org/10.1126/science.348.6237.877-e , May-2015
Significant Media Coverage	Rini M. (Kundan A, Plawsky JL, Wayner PC Jr, et al.) "Synopsis: Staying cool in outer space. Research Highlight of CVB group's article 'Thermocapillary phenomena and performance limitations of a wickless heat pipe in microgravity.' Phys Rev Lett. 2015 Apr 10;114(14):146105. http://dx.doi.org/10.1103/PhysRevLett.114.146105 " Phys Rev Lett. 2015 Apr;114(14). https://physics.aps.org/synopsis-for/10.1103/PhysRevLett.114.146105 , Apr-2015